**Question:**

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

**Reasoning + Answer:**

<think>

Key idea: For low-speed electrons, the kinetic energy relation is K âīĪ (1/2) m v^2, so de Broglie wavelength Î» = h / p = h / (m v) âīĪ (h / m) âī¦ (2K / m).

- Electron rest mass m âīĪ 9.11 ÃĹ 10^-28 g
- 1 eV âīĪ 4.135 ÃĹ 10^-12 erg
- Kinetic energy K = 1 eV = 4.135 ÃĹ 10^-12 erg
- 1 eV corresponds to frequency Î½ = E/h = (4.135 ÃĹ 10^-12 erg) / (6.625 ÃĹ 10^-27 erg Â·s) âīĪ 6.24 ÃĹ 10^14 s^-1
- Wavelength Î» = c / Î½ âīĪ 3.00 ÃĹ 10^10 cm/s / 6.24 ÃĹ 10^14 s^-1 âīĪ 4.81 ÃĹ 10^-5 cm
- Alternatively, de Broglie: h / Î» = p = âī¦ (2 m K), giving Î» = h / âī¦ (2 m K); using K = 4.135 ÃĹ 10^-12 erg and m = 9.11 ÃĹ 10^-28 g yields Î» âīĪ 4.14 ÃĹ 10^-10 m = 4.14 ÃĹ 10^-5